

Max Marchionda

Sr. Principal Software Engineer — Machine Learning

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APPLIED AI · FULL-STACK · SHIPPED IN PRODUCTION

I build AI systems that turn complex data into products people actually use.

Machine learning, agentic AI, and full-stack engineering — from first prototype to production infrastructure serving thousands of users. The right system starts with the right question — I'm most useful when the opportunity is real but the shape of the solution isn't obvious yet.

EXPERIENCE

Sr. Principal Software Engineer — Machine Learning

June 2017 – Present

Optum (UnitedHealth Group) · Minneapolis, MN

Sr. Principal Software Engineer (Feb 2024–Present) → Principal Software Engineer (Jul 2022–Feb 2024) → Senior Software Engineer (Feb 2021–Jul 2022) → Software Engineer II (Jun 2019–Feb 2021) → Data Scientist II (Jun 2018–Jun 2019) → Data Scientist (Jun 2017–Jun 2018)

- Leading engineering for a multi-team, cross-functional initiative within the business segment — partnering directly with business and technology leadership on design and delivery, and building core systems through agentic AI-driven development (AIDLC); backed by a \$50M investment over two years, 8 months in and currently in beta ahead of a full production rollout with measurable business impact in 2027.
- Led and built an AI-powered benefits planning application that leverages machine learning to optimize health benefit configurations, surfacing recommendations through a complex business logic grid evaluated on both a plan and market basis — enabling more strategic, data-driven benefit decision-making at scale.
- Designed and led development of a custom web application that processes competitor ad data using a multi-modal agentic AI workflow, analyzing thousands of ads monthly to automatically surface unique insights — saving over \$2M in annual operational costs.
- Architected and built an internal AI-powered chat experience enabling employees to query 25,000+ developer and data documentation pages, supporting 1,500+ active users and streamlining access to critical technical knowledge.
- Led a cross-functional team in designing and deploying a model discovery and consumption platform at UHG, accelerating AI adoption across 15+ business units and enabling tracking of 450+ AI projects through the full ML lifecycle.
- Contributed to a scalable enterprise ML development platform, optimizing model workflows and infrastructure for production-grade deployment.
- Developed and deployed end-to-end training and scoring pipelines for predicting 20+ chronic and acute conditions (including Atrial Fibrillation and Diabetes) using healthcare claims and clinical data.

WHERE I ADD THE MOST LEVERAGE

Agentic AI Systems — Multi-modal agents that read, reason, and act — production workflows that replace manual analysis at scale.

LLMs · Agentic workflows · AIDLC · Azure OpenAI · Gemini

ML Platforms & Infrastructure — Model discovery, training, and deployment infrastructure that lets whole orgs ship ML responsibly.

MLOps · Azure · GCP · AWS

Full-Stack Product — From data pipeline to polished UI — I own the whole stack so nothing gets lost in translation.

React · Next.js · TypeScript · Node.js

Applied Data Science — Predictive models grounded in real clinical and business outcomes, not just accuracy metrics.

Python · Healthcare claims · Predictive modeling

ALSO BUILDING

Owner, Someday Cards — Built Someday Cards into a \$1M+ revenue TCG business — sourcing, grading, and selling trading cards at scale. Now beginning a new venture: a vendor-centric platform bringing real-time, AI-driven market insights and pricing tools to inventory management. (somedaycards.com)

\$1M+ revenue · E-commerce · Vendor platform · AI-driven pricing

EDUCATION

B.S. Computer Science · B.S. Chemistry — University of Minnesota, Minneapolis, MN (Sept 2012 – May 2017, GPA 3.40)